

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** N,N-Dimethylformamide

**CAS No** 68-12-2

**Synonyms** DMF

**Product Code** 43997

**Address** ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
 Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Acute Dermal Toxicity

Acute Inhalation Toxicity - Vapors

Serious Eye Damage/Eye Irritation

Reproductive Toxicity

Category 4

Category 4

Category 2

Category 1B

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark



Health Hazard

**Signal Word****Danger****Hazard Statements**

H226 - Flammable liquid and vapor

H319 - Causes serious eye irritation

H360 - May damage fertility or the unborn child

H312 + H332 - Harmful in contact with skin or if inhaled

**Precautionary Statements**

P240 - Ground and bond container and receiving equipment

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P312 - Call a POISON CENTER or doctor if you feel unwell

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

**Section 3 - Composition and Information on Ingredients**

| Component         | CAS No  | Weight % |
|-------------------|---------|----------|
| Dimethylformamide | 68-12-2 | >95      |

**Section 4 - First Aid Measures****Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

|                                            |                                                                                                                                                                                                           |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                          |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                      |
| <b>Most important symptoms and effects</b> | Irritating to eyes. Difficulty in breathing. May be harmful if absorbed through skin:<br>Gastrointestinal discomfort: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                           |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Do not use water jetstream.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>).

### Decomposition Temperature

> 350°C

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Thermal decomposition can lead to release of irritating gases and vapors.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component         | Australia                                | New Zealand WEL                                 | ACGIH TLV          | The United Kingdom                                                                                                    | Germany                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------|-------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethylformamide | TWA: 10 ppm<br>TWA: 30 mg/m <sup>3</sup> | TWA: 5 ppm<br>TWA: 15 mg/m <sup>3</sup><br>Skin | TWA: 5 ppm<br>Skin | STEL: 10 ppm 15 min<br>STEL: 30 mg/m <sup>3</sup> 15 min<br>TWA: 5 ppm 8 hr<br>TWA: 15 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 5 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 15 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 5 ppm (8 Stunden). MAK<br>TWA: 15 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 10 ppm<br>Höhepunkt: 30 mg/m <sup>3</sup><br>Haut |

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

| Component         | Australia | New Zealand | European Union | United Kingdom | Germany                                                                                                                                                                                                                                                                                                                    |
|-------------------|-----------|-------------|----------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethylformamide |           |             |                |                | N,N-Methylformamide plus<br>N-Hydroxymethyl-N-methylformamide: 20 mg/L urine (end of shift)<br>N-Acetyl-S-(methylcarbamoyl)-L-cystein: 25 mg/g Creatinine urine (end of shift)<br>N-Acetyl-S-(methylcarbamoyl)-L-cystein: 25 mg/g Creatinine urine (for long-term exposures: at the end of the shift after several shifts) |

### Exposure Controls

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.5 mm          | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene       | < 100 minutes     | 0.45 mm         |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Type A Organic gases and vapours filter Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

|                            |                   |                                  |
|----------------------------|-------------------|----------------------------------|
| Appearance                 | Colorless         |                                  |
| Physical State             | Liquid            |                                  |
| Odor                       | Rotten-egg like   |                                  |
| Odor Threshold             | No data available |                                  |
| pH                         | 6-8 @ 20°C        | 20% aq.sol                       |
| Melting Point/Range        | -61 °C / -77.8 °F |                                  |
| Softening Point            | No data available |                                  |
| Boiling Point/Range        | 153 °C / 307.4 °F |                                  |
| Flash Point                | 58 °C / 136.4 °F  | Method - Abel-Pensky (DIN 51755) |
| Evaporation Rate           | 0.17              | (Butyl Acetate = 1.0)            |
| Flammability (solid,gas)   | Not applicable    | Liquid                           |
| Explosion Limits           | Lower 2.2 vol%    |                                  |
|                            | Upper 16 vol%     |                                  |
| Vapor Pressure             | 4.9 mbar @ 20 °C  |                                  |
| Vapor Density              | 2.5               | (Air = 1.0)                      |
| Specific Gravity / Density | 0.945             | @ 20 °C                          |
| Bulk Density               | Not applicable    | Liquid                           |

|                                                |                          |                                        |
|------------------------------------------------|--------------------------|----------------------------------------|
| <b>Water Solubility</b>                        | Soluble                  |                                        |
| <b>Solubility in other solvents</b>            | No information available |                                        |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                        |
| <b>Component</b>                               | <b>log Pow</b>           |                                        |
| Dimethylformamide                              | -1.028                   |                                        |
| <b>Autoignition Temperature</b>                | 445 °C / 833 °F          |                                        |
| <b>Decomposition Temperature</b>               | > 350°C                  |                                        |
| <b>Viscosity</b>                               | 0.8 mPa.s at 20 °C       |                                        |
| <b>Explosive Properties</b>                    | Not explosive            | explosive air/vapour mixtures possible |
| <b>Oxidizing Properties</b>                    | No information available |                                        |

**Other information**

|                          |                    |
|--------------------------|--------------------|
| <b>Molecular Formula</b> | C3 H7 N O          |
| <b>Molecular Weight</b>  | 73.09              |
| <b>Surface tension</b>   | 36.42 mN/m (25 °C) |

## Section 10 - Stability and Reactivity

|                               |                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>             | None known, based on information available                                                                        |
| <b>Stability</b>              | Stable under normal conditions.                                                                                   |
| <b>Conditions to Avoid</b>    | Incompatible products, Heat, flames and sparks, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b> | Strong oxidizing agents, Halogens, Halogenated compounds, Reducing Agent, . Alkali metals                         |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>).

**Hazardous Polymerization** Hazardous polymerization does not occur.

## Section 11 - Toxicological Information

**Information on Toxicological Effects****Product Information****(a) acute toxicity;**

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Based on available data, the classification criteria are not met |
| <b>Dermal</b>     | Category 4                                                       |
| <b>Inhalation</b> | Category 4                                                       |

| Component         | LD50 Oral          | LD50 Dermal                             | LC50 Inhalation     |
|-------------------|--------------------|-----------------------------------------|---------------------|
| Dimethylformamide | 3040 mg/kg ( Rat ) | 1500 mg/kg (Rabbit)<br>3.2 g/kg ( Rat ) | >5.58 mg/L/4h (Rat) |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;**

|                              |                    |
|------------------------------|--------------------|
| <b>Test species</b>          | rabbit             |
| <b>Observation end point</b> | Irritating to eyes |

**(d) respiratory or skin sensitization;**

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

| Component | Test method | Test species | Study result |
|-----------|-------------|--------------|--------------|
|-----------|-------------|--------------|--------------|

|                                      |                                        |            |                   |
|--------------------------------------|----------------------------------------|------------|-------------------|
| Dimethylformamide<br>68-12-2 ( >95 ) | Guinea Pig Maximisation Test<br>(GPMT) | guinea pig | - non-sensitising |
|--------------------------------------|----------------------------------------|------------|-------------------|

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component         | Australia | New Zealand | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|-------------------|-----------|-------------|-----------------|-------------------|----------|----|----|---------|
| Dimethylformamide |           |             |                 |                   | Group 2A |    |    |         |

(g) reproductive toxicity;  
**Reproductive Effects**  
**Developmental Effects**  
**Teratogenicity**  
 Category 1B  
 Experiments have shown reproductive toxicity effects on laboratory animals  
 May cause harm to the unborn child Developmental effects have occurred in experimental animals  
 Teratogenic effects have occurred in experimental animals.

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** May be harmful if absorbed through skin: Gastrointestinal discomfort: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

### Ecotoxicity effects

| Component         | Freshwater Fish                                                                                                                          | Water Flea           | Freshwater Algae     | Microtox                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|-------------------------------------------------|
| Dimethylformamide | Pimephales promelas:<br>LC50 = 10.6 g/L/96h<br>Onchorhynchus mykiss:<br>LC50 = 9.8 g/L/96h<br>Lepomis macrochirus:<br>LC50 = 6.3 g/L/96h | EC50 = 7500 mg/L/48h | EC50 = 7500 mg/L/96h | EC50 = 2000 mg/L 5 min<br>EC50 = 570 mg/L 240 h |

**Persistence and Degradability**  
**Persistence** Readily biodegradable  
 Persistence is unlikely.

| Component                            | Degradability           |
|--------------------------------------|-------------------------|
| Dimethylformamide<br>68-12-2 ( >95 ) | 100 % (OECD 301E (21d)) |

**Degradation in sewage treatment plant** Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Dimethylformamide | -1.028  | 0.3 - 1.2 L/kg                |

**Mobility** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility but will likely degrade over time: Will likely be mobile in the environment due to its water solubility Highly mobile in soils

### Endocrine Disruptor Information

| Component         | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Dimethylformamide | Group III Chemical                       |                                                  |                                         |

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                             |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.                   |

## Section 14 - Transport Information

### IMDG/IMO

|                             |                       |
|-----------------------------|-----------------------|
| <b>UN-No</b>                | UN2265                |
| <b>Proper Shipping Name</b> | N,N-DIMETHYLFORMAMIDE |
| <b>Hazard Class</b>         | 3                     |
| <b>Packing Group</b>        | III                   |

### ADG

|                             |                       |
|-----------------------------|-----------------------|
| <b>UN-No</b>                | UN2265                |
| <b>Proper Shipping Name</b> | N,N-DIMETHYLFORMAMIDE |
| <b>Hazard Class</b>         | 3                     |
| <b>Packing Group</b>        | III                   |

| Component                            | Hazchem Code |
|--------------------------------------|--------------|
| Dimethylformamide<br>68-12-2 ( >95 ) | 2Y           |

### IATA

|                             |                       |
|-----------------------------|-----------------------|
| <b>UN-No</b>                | UN2265                |
| <b>Proper Shipping Name</b> | N,N-DIMETHYLFORMAMIDE |
| <b>Hazard Class</b>         | 3                     |
| <b>Packing Group</b>        | III                   |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations                      **Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                   | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                              |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethylformamide - 68-12-2 | Schedule 5 listed - in preparations containing 10 per cent or less of dimethylformamide except in silicone rubber mastic containing <=2% of Dimethylformamide<br>Schedule 6 listed - except when included in Schedule 5, or in Silicon rubber mastic containing <=2% of Dimethylformamide |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                   | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-----------------------------|-------------------------------------------------------------|------------------------|
| Dimethylformamide - 68-12-2 | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component         | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Dimethylformamide | X    | X     | 200-679-5 | -      | X    | X   | -    | X     | X    | X    | X     | KE-11411 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|-----------|------------------------------------|------------------------------------------------------------------------|
|-----------|------------------------------------|------------------------------------------------------------------------|

|                             |               |                                 |
|-----------------------------|---------------|---------------------------------|
| Dimethylformamide - 68-12-2 | Annex I - Y42 | Y42 except Halogenated solvents |
|-----------------------------|---------------|---------------------------------|

| Component         | CAS No  | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------|---------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Dimethylformamide | 68-12-2 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

#### Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dimethylformamide | -                                                                   | Use restricted. See item 72. (see link for restriction details)<br>Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details)<br>Use restricted. See item 76. (see link for restriction details) | SVHC Candidate list - (Toxic to Reproduction, Article 57c)                                            |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

#### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**Predicted No Effect Concentration (PNEC)**  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail  
**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

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Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Chemical incident response training.

**Revision Date** 18-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**